

Catch every strip defect where it is born, not where it is shipped

AI surface inspection for stainless strip that detects, classifies and scores every defect at line speed, then routes it to the process that caused it. Built, validated on three public steel datasets, and running live today.

0.764

detection accuracy on unseen steel images (published range 0.70-0.80)

93.7% to 32.5%

clean steel wrongly flagged, before vs after clean-strip training

INR 2.5-4.2 cr

per line, estimated, vs USD 1-3M for a commercial system

~2.2 years

payback for one line at a conservative 1% downgrade rate



[TRY THE LIVE DEMO](#)

surface-vision.github.io

The trained detector runs in your browser.
Upload a strip image to get box, type, confidence and severity. Nothing leaves the page.

1 Problem

Where is value lost?

2 Insights

What did we learn?

3 Solution

What do we build?

4 Implementation

How does JSL deploy it?

5 Impact

What is it worth?

A defect born at the caster is paid for at every later stage, but today it is found at the end of the line

Where stainless surface defects are born, and where they are found without automated inspection

BORN HERE
Inclusions, slivers

BORN HERE
Rolled-in scale, pits,
edge cracks

BORN HERE
Pickling patches, over /
under-pickle

BORN HERE
Scratches, roll marks,
crazing

BORN HERE
Stains, BA marks,
handling scratches

BORN HERE
Found here or by the
customer

Melt & cast

SMS

Hot rolling

HSM

Anneal & pickle

HRAP

Cold rolling

CRM

Final anneal / BA

CRAP / BA

Finish & despatch

Slit, cut

SEEN TODAY

none

SEEN TODAY

none

SEEN TODAY

spot check

SEEN TODAY

spot check

SEEN TODAY

visual sample

SEEN TODAY

visual + claims

Value added keeps rising left to right: melt, rolling, annealing and pickling are all paid for before the coil is graded. "Seen today" is typical practice without automated surface inspection; JSL's own inspection coverage by line is a PO input.

Cost-of-quality lens (PAF): where AI inspection moves the money

	Prevention	Appraisal	Internal failure	External failure
Today	Process fixes happen after complaints	Manual, sampled, inspector-dependent	Downgrade, rework, re-pickle, scrap	Customer claims, returns, lost orders
With AI	Typed defect sent to its owner in minutes	100% of the surface, the same rule on every coil	Caught upstream, before more value is added	Fewer escapes to exposed-finish customers

Output: shift spend from failure costs (right) to appraisal and prevention (left). Cost of poor quality averages ~15% of sales in manufacturing, range 5-35% (IISE).

Sources: AMETEK Surface Vision, Ternium case study (2020); IISE, Measuring the Cost of Quality; ASTM A240 (304: 8.0-10.5% Ni); LME nickel USD 16,402/t and INR 95.8/USD, 22 Sep 2026; JSL FY26 results.

Why it hurts more on stainless

300 m/min

"...it is impossible for the human eye to inspect the whole surface."
AMETEK / Ternium, cold mill

INR 1.26-1.65 lakh

of nickel in every tonne of 304 (8-10.5% Ni). Downgrading it gives away far more value than on carbon steel

2.57 Mt

JSL FY26 sales at INR 1,67,407/t blended realisation, so every 0.1% downgraded is ~2,570 t

Problem statement

How might JSL detect, classify and score every surface defect at line speed, early enough to act, and tell the right process owner what caused it?

The problem is not that defects exist. They are found late and by sampling, and nobody is told which process caused them.

On stainless the surface is the product, so grade, finish and root cause decide which defect matters

Three facts that change the brief

1

The finish cannot be repaired

For mill finishes like 2B and BA, "surface damage such as scratches, grinding marks or spatter cannot be matched by polishing" (ASSDA).

2

JSL is moving toward surface-critical tonnes

Cold rolling grows 2.05 to 2.67 MTPA by FY28. The 300-series mix held margins in Q1 FY27. Exports are 11% (Japan, Korea, Brazil).

3

End markets see the surface

Automotive, railways, metros and white goods drive demand (Q1 FY27 call). BA goes into appliance interiors.

Framework: grade-by-defect risk matrix (which defects to train for first)

Grade family	Slivers, inclusions	Scale, pickling	Scratches, roll marks	Edge cracks	Ridging, roping	Typical exposed use
200 series (Cr-Mn)	M	M	H	M	L	Kitchenware, tubes
300 series (304, 316)	H	M	H	L	L	Appliances, auto, rail
400 series (409, 430)	H	M	M	M	H	Exhaust, white goods
Duplex (2205)	M	H	M	H	L	Process, infrastructure

H / M / L = relative surface risk. Team hypothesis from metallurgy literature (e.g. TiN-cluster slivers in Ti-stabilised grades, ridging in ferritic 430, poor hot ductility in duplex); to be re-scored in P0 against JSL's defect records. Output: 300 series carries the highest risk on exposed finishes and the most nickel per tonne, so it is the pilot grade.

Framework: root-cause routing (condensed fishbone). Each class the model outputs maps to a process owner

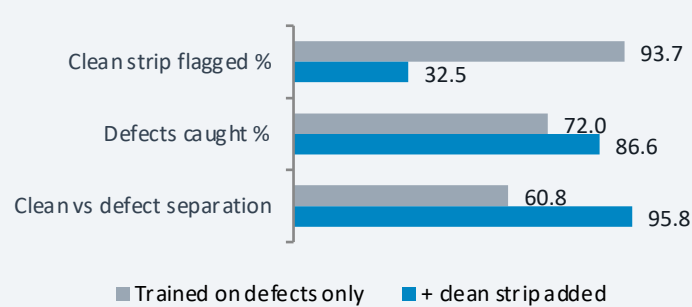
Defect family	Born at	Likely cause (evidence)	Owner	First corrective action
Inclusion, sliver	Caster	Low tundish level, mould-flux entrainment, reoxidation (Leão 2021)	SMS	Hold tundish level, review flux; tag the heat ID
Rolled-in scale, pits	Reheat, HSM	Descaler pressure below spec, blocked nozzles (Steel in Translation 2023)	HSM	Keep descaler above ~180 bar, clear nozzles
Patches, pickling marks	HRAP / CRAP	Uneven scale removal; acid concentration, temperature, line speed	Pickling line	Check acid concentration and speed across width
Scratches, roll marks	CRM, handling	Guides, roller tables, roll surface condition, coil handling	CRM / logistics	Inspect guides, trigger roll change
Edge cracks	HSM	Edge cooling, low hot ductility (duplex, ferritic)	HSM	Edge heating and trim practice; needs edge cameras

Sources: ASSDA, 2B/2D/BA finishes; JSL Q1 FY27 earnings call and results (Aug 2026); JSL cold-rolling expansion to 2.67 MTPA; Steel in Translation 2023 (rolled-in scale); Leão et al. 2021 (tundish slivers). Matrix: team hypothesis.

Classify, don't just alarm: a typed defect becomes a work order for a named owner. The pilot belongs on a 300-series cold-rolled line.

Our Round 1 build produced four findings, and each one changed the design

1 A model that has never seen clean steel raises false alarms on it



On 505 certified-clean steel images, a model trained only on defects flagged 93.7% as defective. Adding real clean strip to training cut that to 32.5%, and accuracy went up, not down. Training on clean strip alone fails: it stops finding defects.

So we: train on JSL's own clean and defective strip before any alarm is trusted.

2 A bigger model or sharper images did not help

Option tested	Accuracy	Extra cost	Verdict
Compact model	0.752	-	Chosen
Compact model + clean strip	0.764	none	Ready
Model 3.7x larger	0.734	3.5x compute	No gain
4x sharper input images	0.734	4x pixels	No gain
Extra processing per image	0.006 lower	2.4x slower	Rejected

Accuracy on held-out images. The gap between compact and large models is within noise.

So we: keep the model compact and spend the budget on camera coverage instead.

3 Line speed, not the model, sets the hardware bill

~18

processors needed to see every 0.2 mm of a 1.28 m wide strip moving at 250 m/min

1

processor is enough if images are coarsened to 1.28 mm, but fine scratches and crazing then disappear

So we: fix the image resolution with JSL at the start, and size the hardware from it.

4 Operators only act on alarms they can trust

68% lower

confidence error after calibration. Before, an alarm marked "60% sure" was right 84% of the time; now stated confidence is within ~5 points of reality

0 of 3,200

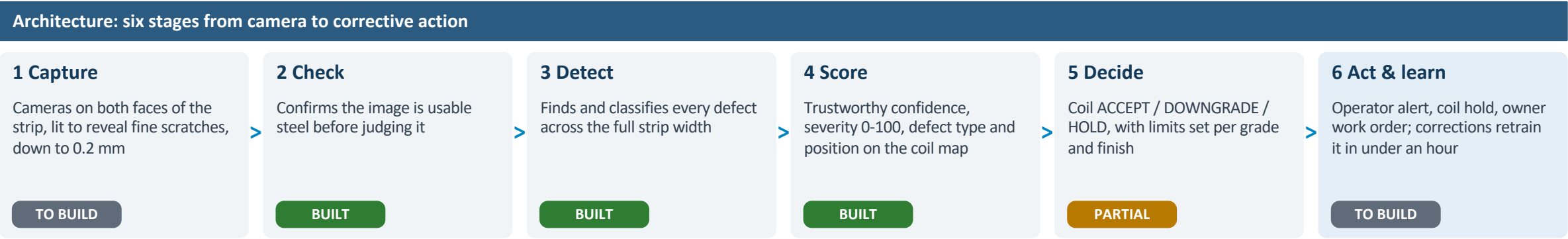
real steel images wrongly rejected by our "is this steel?" check, which stops the system judging anything else (unchecked, it called a plain white square a defect)

So we: give every alarm a trustworthy confidence and a "this is steel" check.

All numbers measured by us on three public steel-surface datasets, on images the model never saw in training.

The model is the cheap part. JSL's own strip, the resolution spec, and trustworthy confidence decide whether the system works.

Detect on the strip, decide at the coil, route to the owner, and learn from every operator correction



What is new beyond a standard detector

- 1

Trained on clean strip, not just defects
Fixes the false-alarm problem that kills most pilots. Measured: 93.7% to 32.5%.
- 2

Root-cause routing
Each class maps to an owner (slide 3), so an alarm becomes a work order, not just a red light.
- 3

Catches defects it was never taught
A second check flags unusual surface patterns that were never labelled. Proposed.
- 4

Learns from JSL's unlabelled coils
Published steel research shows unlabelled images train as well as the standard approach, so JSL's image archive becomes an asset.

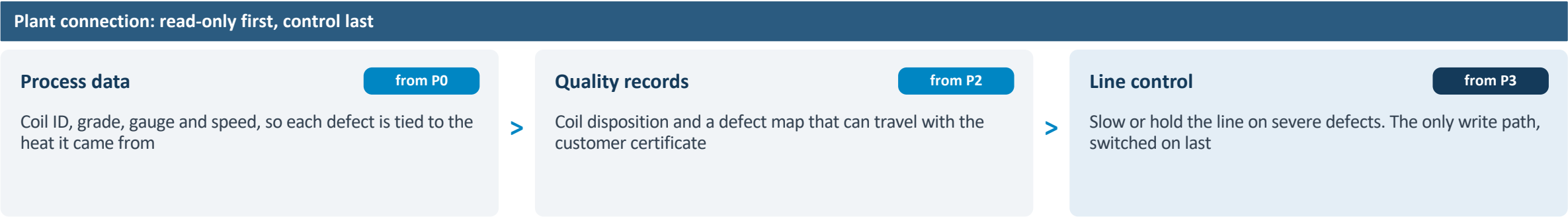
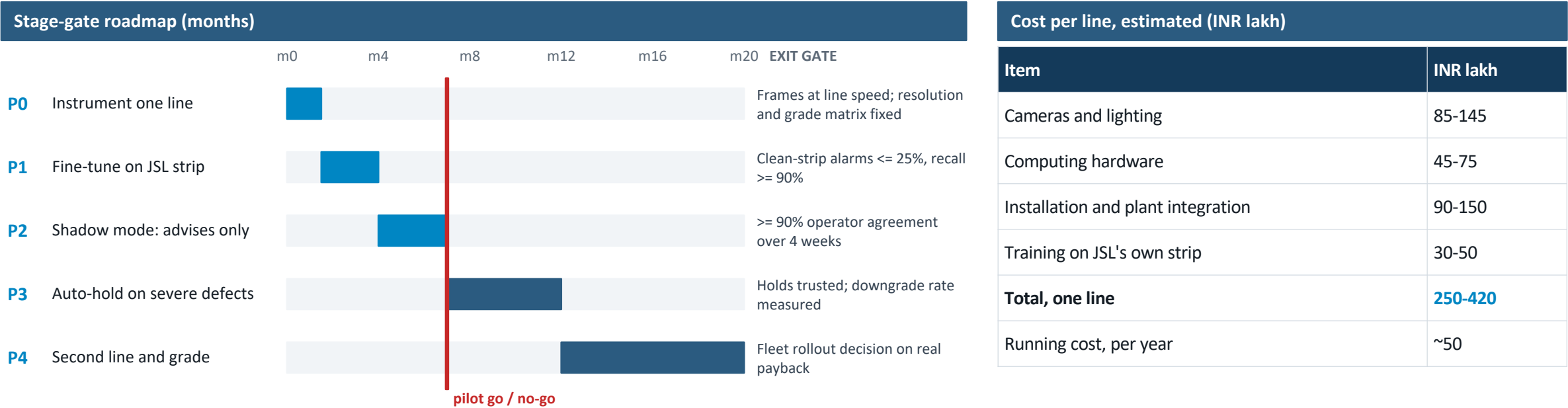
Framework: build vs buy (options scored against JSL's needs)

	Capex per line	Proven at line speed	Tuned to JSL grades	Root-cause routing	JSL owns data and model	Time to first line
Commercial (ISRA, AMETEK) USD 1-3M (unverified)						
Fully in-house INR 2-3 cr (est.)						
Hybrid: buy optics, own model INR 2.5-4.2 cr (est.)						
strong partial weak Output: hybrid wins on 4 of 6 criteria and is never weak.						

BUILT = working in our prototype today, live at surface-vision.github.io. Unlabelled-learning result: J. Mater. Inf. 2025. Commercial price range: industry figure, unverified.

Buy proven optics, own the model. JSL gets vendor-grade capture with a detector trained on its own grades and wired to its own process owners.

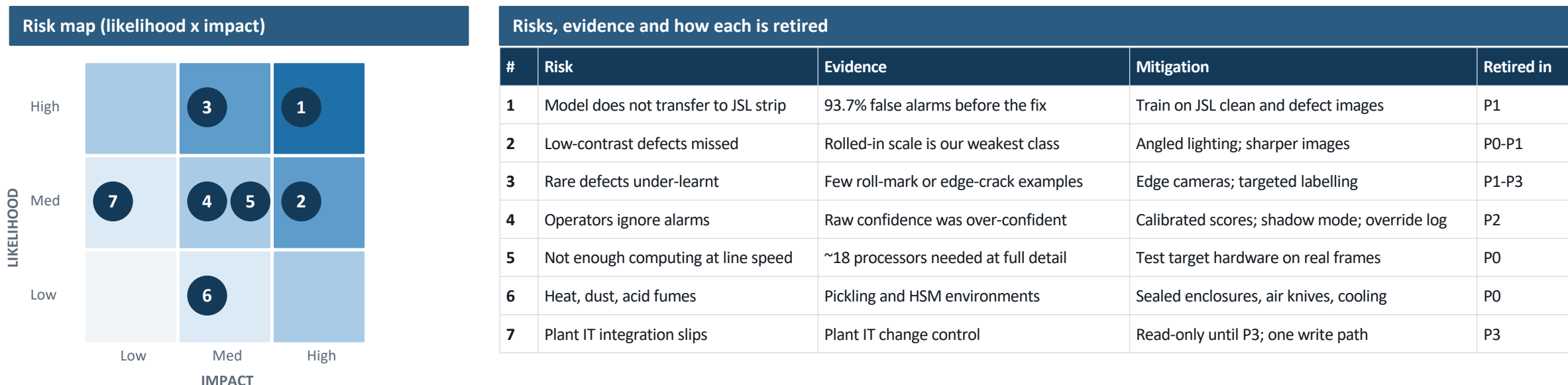
Seven months to shadow mode on one line, and nothing touches the mill until trust is earned



Costs are team estimates for a 1.3-1.6 m line inspected on both faces, at INR 95.8/USD; to be replaced by vendor quotes in P0.

Commit INR 2.5-4.2 cr and one line for seven months. The month-7 go/no-go is decided on JSL's own alarm and recall numbers.

Every high risk is retired in P0-P2, before the system is allowed to slow the mill



Evidence from our own tests on public steel datasets. KPI targets are proposed, to be agreed with JSL Quality in P0.

Accuracy, speed and trust are already proven off-site. What only JSL's line can prove is measured in shadow mode first.

At a conservative 1% downgrade rate one line pays back in ~2.2 years; the fleet protects ~INR 13 cr a year

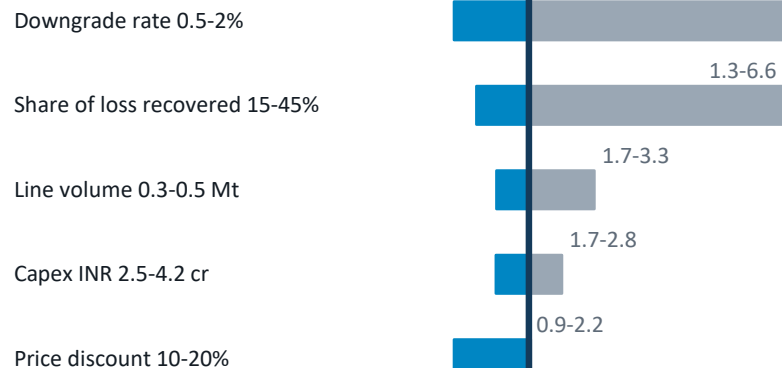
Loss pool per line, INR cr/yr

0.4 Mt x downgrade rate x price discount x INR 1,67,407/t

Downgraded	10% off	15% off	20% off
0.5% of volume	3.3	5.0	6.7
1.0% of volume	6.7	10.0	13.4
2.0% of volume	13.4	20.1	26.8

Outlined: base case, 6.7 cr pool. Recovering 30% = INR 2.0 cr/yr, 1.5 cr net of run cost. Fleet at the same rates (2.57 Mt, ~6.4 lines): INR 43 cr pool, INR 12.9 cr/yr recovered.

What moves payback (years, base 2.2)



Downgrade rate and recovery share dominate. Neither is public, so both are measured in P2-P3 before fleet spend.

Base-case payback

Hybrid (ours)

2.2 yr

Commercial system

12.7 yr

+INR 2.1 cr 5-year NPV per line at 12%

>= 0.11% of line revenue lost to downgrade (rate x discount) is enough for a 2-year payback

Impact beyond the P&L

Coverage

100% of both faces, the same rule on every coil, instead of sampled grading

Customers

Defect map per coil for auto, rail and appliance buyers; fewer claims on exposed finishes

Process

Typed defects tied to heat and coil IDs give the SMS and HSM root-cause data they lack today

Safety and assets

Severe-defect holds guard against strip breaks, the Ternium business case

Sustainability

Less remelt and re-pickling of downgraded coils saves energy, acid and nickel

People

"AI will not replace; it will instead renew and re-energise." Inspectors move to decisions (JSL MD)

WHAT WE ASK OF JSL

- One 300-series cold-rolled line for 7 months
- INR 2.5-4.2 cr pilot budget
- Downgrade rate by line and grade from the MIS
- A metallurgist at 20% to sign off root causes

In return: measured downgrade rate, payback and a go / no-go at month 7

Inputs: JSL FY26 (INR 1,67,407/t; 25,65,902 t). Swept assumptions: 0.4 Mt line, 30% recovered, INR 3.35 cr capex, INR 0.5 cr/yr run cost, 12% rate. Commercial: USD 2M at INR 95.8. MD quote: ET, 16 Jul 2024.

Recommendation: approve a 7-month pilot on one 300-series cold-rolled line. The number that decides fleet rollout is JSL's own downgrade rate.